

Raymond Song CMU-educated Robotics/RL/ML Researcher [linkedin.com/in/rysong](https://www.linkedin.com/in/rysong)
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TECHNICAL SKILLS

Python, C++, PyTorch, JAX, Reinforcement Learning (RL), Sim-to-Real Transfer, System Integration, Simulation(Isaac Sim MuJoCo, Gazebo), Physical Hardware, Locomotion, Manipulation, Machine Learning, Distributed Training, Autonomous Vehicles, SLAM, Computer Vision, ROS/ROS2, DDS, AWS, CUDA, Docker, Linux, CMake, Git, Writing Documentation

EXPERIENCE

Carnegie Mellon University | Robotics Researcher (Advised by Professor Jeff Schneider) **2024 - Present**

- Developed and published **Reinforcement Learning** algorithm in **PyTorch** that extends PPO with teacher-guided exploration to navigate complex off-road driving tasks, distilling state-only policies into perception-based control policies
- Integrated end-to-end perception-based control and GNN RL policies into **C++ ROS** autonomy stack via LibTorch and CUDA for real-time execution on a 1000kg **physical robot**, running up to 50Hz despite on-board compute constraints
- Deployed RL policy zero-shot in off-road environment, achieving 100% **real-world** obstacle avoidance at 10MPH, by closing the **Sim-to-Real** gap with domain randomization, high-fidelity simulation, and vision foundation models
- Designed large-scale simulation training and hardware-in-the-loop (HIL) evaluation pipelines, reward functions, and tuned hyperparameters, outperforming PPO, SAC, Imitation Learning (Dagger, GAIL), and MPC/MPPI baselines
- Leveraged PyTorch profiler to isolate and eliminate bottlenecks to achieve 5x training speedup and save GPU compute
- Devised flexible parameter tuning interface with dynamic reconfiguration and ROS plugins for broad code reusability

AI Racing Tech (\$1.5M Autonomous Racecar) | Various roles incl. Data Engineer, Deployment & Testing Lead **2023 - 2025**

- Orchestrated field tests for autonomous Indy Car racing at 180MPH from system bring-up and trackside safety with 0 crashes in 2 years to robot deployment pipelines, hands-on triage of on-site issues, saving deployment cycles and money
- Lead feature integration and safety-critical code review, working across perception, planning, and control teams to debug the full stack from high-level software to low-level hardware for seamless transitions from development to deployment
- Created lightweight **C++ ROS2** fault detection telemetry tooling to ensure safety in a fast-paced and high-stakes environment, streamlining data-driven feedback loops to drive autonomy improvements increasing speeds by 40MPH

TritonAI Autonomous Racing (*Autonomous Go-kart*) | Lead Robotics Engineer **2023 - 2024**

- Lead team of 5 engineers to create autonomous go-kart stack via custom ROS2 Docker containers for Nvidia Jetson platforms, YOLOv8 detection for obstacles, GPS localization, and pure pursuit controller for speeds at 16MPH
 - Built electronic control unit substitute with VESC and STM32 for 200Hz low-latency control for steering and throttle
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PUBLICATIONS: *Several publications, only some are listed below*

- **R. Song***, Z. Wu*, V. Mundheda*, L. E. Navarro-Serment, C. Schoenborn, J. Schneider, “TADPO: Reinforcement Learning Goes Off-Road”, ICRA 2026 (**Selected Oral Presentation, Top 11% of Papers**)
 - T. Gupta*, A. Verma*, **R. Song**, D. Guttendorf, C. Igoe, L. E. Navarro-Serment, J. Schneider, “Efficient Active Search via Amortized Path-Integral Policies”, ICRA 2026
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OPEN SOURCE CONTRIBUTIONS (2020 - Present): *Several projects, only some are listed below*

TADPO: Teacher Action Distillation with Policy Optimization

- Teacher-student RL algorithm that integrates expert action distillation with on-policy learning to tackle poor exploration without explicit reward shaping while maintaining training stability and avoiding behavior cloning’s OOD problems

Chest X-Ray Expert Radiologist Report Generator

- Fine-tuned VLM (LLaVA) with LORA modules and context-embedded prompts to generate expert radiologist chest X-rays reports to help radiologists swiftly and accurately identify illnesses like pneumonia or pneumothorax
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EDUCATION Carnegie Mellon University: Master of Science, Robotics | UC San Diego: BSc, Data Science (**Top 4%**)

Relevant Coursework: Robot Learning, Manipulation, Robot Safety, Deep Learning (CNN, Transformers, VLA), SLAM

PUBLIC TECH TALKS Master’s Thesis: Real-World Off-Road Driving with RL (CMU), TADPO (ICRA 2026)

INTERESTS Reading novels by Dostoevsky | Getting lost in art museums | Steaks after strength training